Hence, by taking vertical strips, we get

$$A = 2\int_0^a y\,dx = 2\int_0^a 2\sqrt{a}\sqrt{x}\,dx = 4\sqrt{a}\left[\frac{2}{3}x^{\frac{3}{2}}\right]_0^a$$

$$= 4\sqrt{a}\times\frac{2}{3}a^{\frac{3}{2}} = \frac{8a^2}{3}.$$

Note

Viewing in the positive direction of x-axis, and making horizontal strips (see Fig. 9.19), we get

$$A = 2\int_0^{2a}(a-x)dy = 2\int_0^{2a}\left(a-\frac{y^2}{4a}\right)dy$$

$$= 2\left(ay-\frac{y^3}{12a}\right)_0^{2a} = 2\left(2a^2-\frac{8a^3}{12a}\right) = \frac{8a^2}{3}.$$

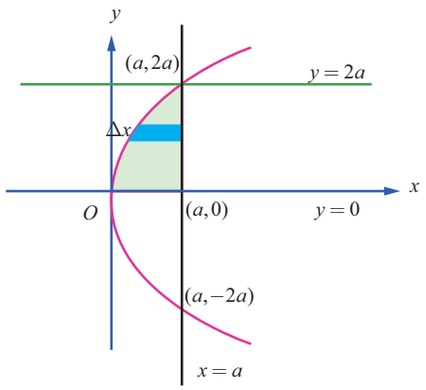


Fig. 9.19

Note

It is quite interesting to note that the above area is equal to two-thirds the base (latus-rectum) times the height (the distance between the focus and the vertex). This verifies Archimedes' formula for areas of parabolic arches which states that the area under a parabolic arch is two-thirds the area of the rectangle having base of the arch as length and height of the arch as the breadth. It is also equal to four-thirds the area of the triangle with base (latus-rectum) and height (the distance between the focus and the vertex).

Example 9.51

Find the area of the region bounded by the y-axis and the parabola $x = 5-4y-y^2$.

Solution

The equation of the parabola is $(y+2)^2 = -(x-9)$. The parabola crosses the y-axis at $(0,-5)$ and $(0,1)$. The vertex is at $(9,-2)$ and the axis of the parabola is $y=-2$. The required area is sketched as in Fig. 9.20.

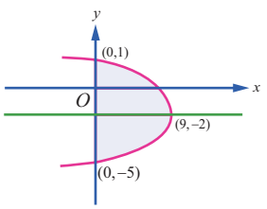


Fig. 9.20

Viewing in the positive direction of $x-$ axis, and making horizontal strips, the required area A is given by

$$A = \int_{-5}^{1} x\,dy = \int_{-5}^{1}(5-4y-y^2)dy = \left[5y-2y^2-\frac{y^3}{3}\right]_{-5}^{1} = \frac{8}{3}-\left(-\frac{100}{3}\right) = 36.$$

Note

As in the previous problem, we again verify Archimedes' formula that the area of the parabolic arch is equal to two-thirds the base times the height.

Example 9.52

Find the area of the region bounded by $x-$ axis, the sine curve $y = \sin x$, the lines $x = 0$ and $x = 2\pi$.

Solution

The required area is sketched in Fig. 9.21. One portion of the region lies above the $x-$ axis between $x = 0$ and $x = \pi$, and the other portion lies below $x-$ axis between $x = \pi$ and $x = 2\pi$. So, the required area is given by

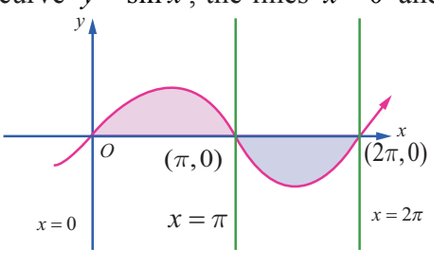


Fig. 9.21

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